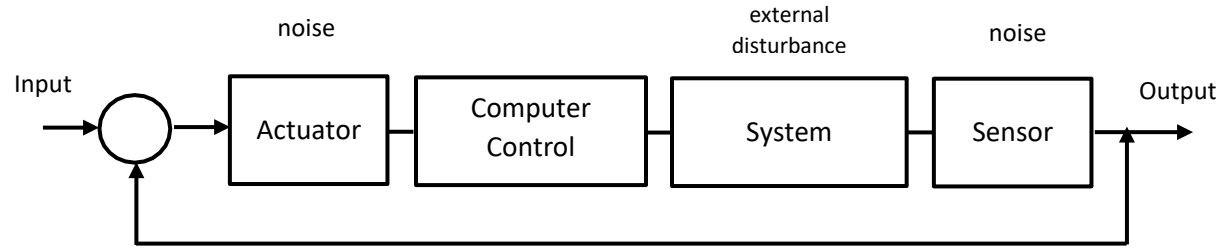
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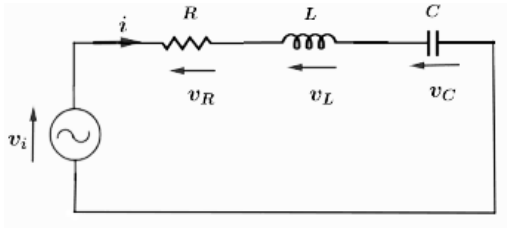
Part 2 : Tutorial 1 – Intro & System Modelling

Q1 : Identify any three feedback systems that you encounter in your everyday environment. For each system identify the sensing mechanism, actuation mechanism and control law. Describe the uncertainty with respect to which the feedback system provides robustness and/or dynamics that are changed using feedback**.**

Q2 : Consider balancing our self on one foot with our eyes closed for 15sec as a balance system. Using the block diagram below as a guide, describe the control system responsible for preventing you from falling.



Q3 : Derive the Transfer Function $\left(\frac{I(s)}{V_i(s)}\right)$ – in the Laplace Transform domain - of the RLC network as shown below :



RLC Network (2nd Order System)

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Part 2 : Tutorial 2 – Analogue Control System

Q1 : Describe all the possible outcome of a 2nd order system (with the aid of diagrams in the s-domain) when the input of the system is a unit step.

Q2 : A servomechanism controls the angular position (θ) of a mass through a command signal ($\boldsymbol{u}$). The transfer function in s-domain is given by :

$$\frac{\theta(s)}{u(s)} = \frac{344}{s^2 + 25s + 344}$$

Determine the transient response characteristics (percentage peak overshoot, rise time, settling time, and peak time) of the open loop system?

Q3 : The Open-Loop System Model of a 2nd Order System is $\frac{1}{s^2+6s+5}$. Users find this system to be quite sluggish in the response to a step input. The resident engineer suggest that the response of the system can be improved by creating a unity gain negative feedback control system and adding a simple proportional controller with a gain of ***K***. Calculate the value of ***K*** if you want the step response to be critically damped ($\zeta = 1$)

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Part 2 : Tutorial 3 – Intro to Digital Control System

Q1 : Use Initial Value Theorem to calculate the value of *x(0)* for the following system transfer functions (Z-Transform) :

i) $X(z) = \frac{z^2+z+1}{(z+2)(z+1)}$

ii) $X(z) = \frac{z+4}{(z+2)(z+1)}$

Q2 : Use Final Value Theorem to calculate the value of $x(\infty)$ for the following system transfer functions (Z-Transform) :

i) $X(z) = \frac{z^2}{(z-1)(z-0.3)}$

ii) $X(z) = \frac{z+1}{3(z-1)(z+0.4)}$

Q3 : Consider a system described by the following difference equations :

$$y(k) + 2.1y(k-1) + 1.4y(k-2) + 0.32y(k-3) = x(k)$$

where $y(k)$, $x(k)$ are the output/input of the system.

If the input to this system is an impulse function, find the resultant output, *Y(z)*.

Q4 : A mechanical lifting system (with the following transfer function) is shown below :

$$T.F. = \frac{Y(z)}{X(z)} = \frac{1}{z(z+8)(z+1)}$$

If the input to this system is a sampled step function, find the difference equation that you will need to implement for a software simulation of this system.

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Part 2 : Tutorial 4 – Transient Analysis & Stability

Q1 : The transfer function of an open-loop chemical plant, G(z), is given here. A controller function, C(z), is also given. Design a closed-loop unity gain negative feedback system for this plant and the controller. Find the Controller Gain. K, for the system to have a damping ratio of 0.3, a natural undamped frequency of 70 rad/s and a sampling rate of 20 samples/sec.

$$G(z) = \frac{1}{z^2 + 0.686z + 0.002} \qquad C(z) = \frac{K}{200}$$

Q2 : The system characteristic equation of the mechanical system is given as :

$$P(z) = z^2 - 1.321z + 0.497$$

Find the damping ratio, ζ, and undamped natural frequency, ω_n for this system when the sampling period is 0.2s.

Q3 : Consider the following open-loop transfer function in the z-domain

$$G(z) = \frac{1}{(z-1)(z-0.3)}$$

With *T = 0.1s*, design the closed-loop unity gain negative feedback system that includes a proportional controller. Find the proportional gain, ***K***, to achieve the followings:

i) A damped natural frequency of 5 rad/s.
ii) A settling time of 2s.

You may assume the following relationship between the damped natural frequency (ω_d) and undamped natural frequency (ω_n) :

$$\omega_d = \omega_n\sqrt{1-\delta^2}$$

For Question 1 to 3. You may assume the standard characteristic equation for a second order system is (either version can be used) :

$$z^2 - 2\cos\left(\omega_n T\sqrt{1-\delta^2}\right)e^{-\delta\omega_n T}z + e^{-2\delta\omega_n T}$$

$$z^2 - 2\cos(\omega_d T)e^{-\delta\omega_n T}z + e^{-2\delta\omega_n T}$$

Q4 : Examine the stability of the characteristic equation given by :

$$P(z) = (z-1)(z-0.8)(z+0.5)$$

Plot the poles on the z-plane and comments on the positions of the poles and how they affect the stability of a system.

Apply Jury's stability test and comment on the stability of the system.

Q5 : Consider the system described by :

$$y(k) + 2.1y(k-1) + 0.4y(k-2) = x(k) + x(k-1)$$

where $y(k)$, $x(k)$ are output/input of the system. Determine the stability using Jury's stability test.